This is, when individual propensities are manifesting themselves. In evoluation we can see this in evolution in the theory of Evolutionarily Stable Strategies (ESS).

In infectious diseases we have the problem of when and if a few infected individuals can cause an epidemic. An ESS is characterised by that when a strategy is adopted by a majority of a population no rare mutant is able to invade that population. Here we see how the properties of few individuals are having crucial importance of the fate of the many. Or in terms of a dynamic system, is the equlibrium state of the system invasible or susceptibility to change. Here is where dynamical systems theory meets modalities such as possibility and necessity.

> A 'strategy' is a behavioural phenotype; i.e., it is a specification of what an individual will do in any situation in which it may find itself. An ESS is a strategy such that, if all members of a population adopt it, then no mutant strategy could invide the population under the influence of natural selection.

Causation is causal dependence with transitivity (Lewis, 1973, p. 563). Assume that we have a population growing according to some law f and that its population changes from N_t one year to N_{t+x} the year after.

$$N_{t+1} = N_t(1 + f(x, N_t))$$

Further, let AA the dominating genotype in the population, i.e. its wild type. This A gene manifests itself on some trait called x. In particular, let the AA have the specific value of $x = \hat{x}$, and that the growth rate is affected by the value of x

Consider the following series of events:

M *A rare mutation occurs*
Assume that the A gene occasionally mutates $A \to a$ and that the bearers of the allele a (Aa) have a phenotype that deviates from x. Also, if the frequence of the a allele q is small in comparison with that of the A allele which is $p = 1 - q$ and that, according to Hardy-Weinberg law the frequencies of AA is $p^2 \approx 1$, Aa is $2pq \ll 1$ and aa is negligible $q^2 \approx 0$.

P *The muation has a phenotypic effect*
Assume that the mutants with genotype (Aa) have a modified trait where $x = \hat{x} + \delta x$

References

Lewis, David, "Causation," *The journal of philosophy* **70**, pp. 556-567, Seventieth Annual Meeting of the American Philosophical Association Eastern Division (Oct. 11, 1973) (1973). https://www.jstor.org/stable/2025310.